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CO1 – AT1 - APPLICATION BASED QUESTIONS

Course Code – DSA0308

Course Name – Natural Language Processing

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Question 1 Regular Expressions - Resume Information Extraction.

Answer:

- Store the resumes in a text format and import python built-in re module to search and extract information.
- To extract candidate's name use re.search() with a suitable pattern to identify the text labe Name:.
- To identify Email Addresses use a regular expression that matches the username, the @ symbol, the domain name, and the domain extension.
- To extract mobile numbers use a pattern that matches a 10-digit Indian mobile number beginning with a digit from 6 to 9.
- To detect Technical Skills define a list of required skills, such as Python, Java, SQL, Machine Learning, and NLP. Use re.findall() to identify which skills appear in the resume.
- To extract Years of Experience search for phrases such as 2 years of experience or 3 years and extract the numerical value.
- Generate a profile summary display the extracted name, email address, mobile number, years of experience, and technical skills in a structured format.
- Shortlist candidates who have at least two years of experience and possess Python skills. If both conditions are satisfied, mark the candidate as eligible otherwise mark the candidate as not eligible.
- Result**:** The system automatically extracts important candidate information from resumes and identifies candidates who satisfy the minimum eligibility criteria.

Question 2 Regular Expressions – Product Search System.

Answer:

- Create a product list store the names of available products in a Python list.
- Get the search keyword ask the user to enter the keyword they want to search for.
- Use the pattern ^keyword$ to find products whose entire names match the given keyword.
- Perform prefix-based search - use the pattern ^keyword to find products whose names begin with the specified keyword. perform suffix-based Search -use the pattern keyword$ to find products whose names end with the specified keyword.
- Perform partial Keyword search for the keyword without start or end anchors. This finds products containing the keyword anywhere in their names.
- Perform case-insensitive Search use the re.IGNORECASE flag so that uppercase and lowercase letters are treated equally.
- Dispplay the matching products compare the search pattern with every product name and display all products that match the selected search type.
- Generate a search report count the matching products using the number of results obtained. Display the search keyword, search type, matching product names, and total number of matches.
- Repeat the search allow the user to perform multiple searches until they choose to exit the system.
- Result: The system provides flexible product searching through exact, prefix, suffix, partial, and case-insensitive matching and displays the total number of matching products for each search.

Question 3 Regular Expressions – University Registration System

Answer:

- The university registration validation system can be developed using Python's re module to verify student information before course enrollment.
- Cbtain the student's register number, institutional email address, course code, semester, and mobile number.
- Use a regular expression to check whether the register number follows the required format. For example, two digits followed by three uppercase letters and three digits.
- Use a pattern to verify the email format and ensure that the address ends with the university's specified domain, such as @university.edu.
- Check whether the course code follows the prescribed format, such as three uppercase letters followed by three digits.
- Verify that the semester is a valid number within the university's permitted range, such as 1 to 8.

- Use a regular expression to check whether the mobile number contains exactly 10 digits and begins with a digit from 6 to 9.
- Check each field separately and display an appropriate message, such as Valid or Invalid, depending on whether it matches the required pattern.
- Check the validation status of all fields. If every field is valid, display "Registration Successful". If any field is invalid, display "Registration Failed" and ask the student to correct the invalid information.
- Result**:** The system validates the student's registration details, displays individual validation messages, and determines whether the registration is successful.